

WEEK 6 PROGRESS UPDATE

Multilingual Speech-to-Intent Detection System

Research Progress Report

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Reporting Period	Week 6	Status	Ongoing — SLM Candidate Survey

1. Purpose of This Report

Following the Week 5 deep-dive into the MASSIVE dataset's intent taxonomy and its distribution across the five Indian locales, this week's work turned from the data side of the project to the model side. With a clear picture of what training data is available and where the class imbalance sits, I spent Week 6 surveying candidate Small Language Models (SLMs) — both text-based and speech/audio-based — that could replace or sit alongside the Whisper → MuRIL pipeline used so far. The goal of this report is to record, in one place, what each candidate model is, how it is positioned with respect to Indian-language support, and how the candidates group into a short list worth benchmarking on the pooled Hindi, Kannada, Malayalam, Tamil and Telugu MASSIVE subset.

2. What Is a Small Language Model?

A Small Language Model (SLM) is a language model built with a relatively small number of parameters compared with large language models (LLMs), aiming to provide useful language-understanding and generation capability while needing less memory, less compute and less inference time. There is no universally fixed parameter cutoff that defines an SLM — in practice, models ranging from a few hundred million parameters up to a few billion parameters are commonly treated as small.

Characteristics that matter for this project:

- Fewer parameters than general-purpose LLMs, with correspondingly lower memory and compute needs
- Faster inference and lower deployment cost — relevant if the pipeline is ever meant to run on-device
- Easier to fine-tune for a single, narrow task such as intent classification
- Can be specialized for a particular language or application rather than staying fully general-purpose

Because the intent-classification stage of this project is restricted to a fixed, predefined set of 60 intents (the MASSIVE taxonomy covered in the Week 5 report), a very large general-purpose model is not strictly necessary — a smaller model fine-tuned specifically for this classification task is a reasonable starting hypothesis. For example, the input “Set an alarm for 7 AM tomorrow” should map to the single output label `alarm_set`.

3. Two Categories of Models Relevant to the Project

For the purposes of this survey, the candidate models split cleanly into two categories depending on what kind of input they accept.

Category 1 — Text-based SLMs

Indian-language text → Text SLM → Intent

These models take text as input and are useful downstream of a speech-recognition system that has already converted speech into text.

Category 2 — Audio/speech-capable models

Indian-language speech → Speech/Audio Model → Text or language understanding

Some of these models perform only speech-to-text (conventional ASR), while others are audio-language models that reason about spoken input more directly.

4. Text-Based Small Language Models Considered

4.1 Sarvam 2B

Sarvam 2B is an open-source, approximately 2-billion-parameter Indic language model from Sarvam AI, trained from scratch with Indian-language processing specifically in mind. Sarvam positions it as a foundational small Indic LLM covering 10 Indian languages, which makes it the one candidate in this list that is India-focused rather than generally multilingual. Its natural role in the project would be as the intent-understanding component after speech has already been converted to text — for example, a Telugu ASR transcript such as “■■■■■ ■■■■ 7 ■■■■■■ ■■■■■■ ■■■■■■” being classified directly to `alarm_set`.

4.2 Qwen2.5 Small Models

The Qwen2.5 family offers several small sizes — roughly 0.5B, 1.5B and 3B parameters — which makes it useful for studying how intent-classification performance scales with model size while holding the architecture fixed. Qwen2.5 is multilingual but not specifically Indian-language-specialized, so its performance on Hindi, Kannada, Malayalam, Tamil and Telugu would need to be measured experimentally rather than assumed. It is best treated as a general multilingual baseline to compare against an India-focused model like Sarvam 2B.

4.3 Gemma Small Models

Google's Gemma family includes small variants in roughly the 1B–4B range (Gemma 3 1B and Gemma 3 4B). Gemma is multilingual but not India-specific, so — as with Qwen — its Indian-language performance is an open question rather than a given. It serves as a second general multilingual baseline alongside Qwen and Sarvam 2B.

4.4 SmolLM2

SmolLM2 is built specifically around very small parameter counts — approximately 135M, 360M and 1.7B variants — which makes it the natural candidate for asking how small a model can get while still doing something useful with language. SmolLM2 is multilingual but not Indian-language-specialized, so its value here is mainly in probing the research question of how far parameter count can be reduced before Indian-language intent-classification accuracy degrades — relevant if the project ends up caring about edge or resource-constrained deployment.

4.5 Llama 3.2 1B

Meta's Llama 3.2 family includes a 1-billion-parameter text model. Like Qwen, Gemma and SmolLM2, it is multilingual rather than Indian-language-focused, so it rounds out the set of general-purpose baselines against which Sarvam 2B's India-specific training can be compared on the MASSIVE Indian-language subset.

5. Comparison of Text-Based SLMs

Model	Approx. Size	Primary Focus	Indian-Language Orientation	Audio Input
Sarvam 2B	~2B	Indic language understanding	India-focused	No
Qwen2.5 0.5B	~0.5B	Multilingual	General multilingual	No
Qwen2.5 1.5B	~1.5B	Multilingual	General multilingual	No
Qwen2.5 3B	~3B	Multilingual	General multilingual	No
Gemma 3 1B	~1B	Multilingual	General multilingual	No
Gemma 3 4B	~4B	Multilingual	General multilingual	No
SmolLM2 135M	~135M	Very small LM	General multilingual	No
SmolLM2 360M	~360M	Very small LM	General multilingual	No
SmolLM2 1.7B	~1.7B	Small LM	General multilingual	No
Llama 3.2 1B	~1B	Multilingual	General multilingual	No

The clearest split in this table is between Sarvam 2B, the single India-focused SLM, and everything else, which are general multilingual models whose Indian-language behaviour still needs to be verified rather than assumed just because they are described as “multilingual.”

6. Audio/Speech-Based Models Considered

Because the end goal is a speech-intent system and a conventional text SLM cannot read a raw audio waveform, this part of the survey looked at models that sit on the speech side of the pipeline.

6.1 AI4Bharat IndicConformer-600M

IndicConformer-600M-Multi is a multilingual, Conformer-based ASR model built for Indian languages, with roughly 600M parameters and support for 22 Indian languages — including all five that this project needs (Hindi, Kannada, Malayalam, Tamil, Telugu). Its combination of a relatively small size, an Indian-language focus and full coverage of the target languages makes it the strongest candidate for the speech-recognition stage of a two-stage pipeline: speech in, Indian-language text out, which can then be handed to a text SLM such as Sarvam 2B for intent classification.

6.2 Sarvam Saaras

Saaras is Sarvam AI's speech-to-text family, currently supporting 23 languages in total — 22 Indian languages plus English, again covering all five target languages. Beyond plain transcription, Saaras supports translation, verbatim transcription, transliteration and code-mixed output, along with automatic language detection and tolerance for Indian-language accents and code-mixed speech. It is best classified as an Indian-language speech-to-text model rather than a text SLM — its role would be the recognition stage, feeding a separate intent classifier, not the classification itself.

6.3 Shuka v1

Shuka v1 is the most architecturally distinct candidate: an open-source Audio Language Model that takes Indian-language voice input directly and produces text output, combining an audio encoder (Saaras v1) with Meta's Llama 3 8B Instruct as the decoder through a small ~60M-parameter projector. It has been tested across eleven languages including Hindi, Kannada, Malayalam, Tamil and Telugu, having been fine-tuned on English and Hindi data with the multilingual audio encoder enabling zero-shot performance on the other tested languages. The important caveat is that, because its decoder is Llama 3 8B, Shuka should not be described as a small model by total parameter count — it is more accurately an audio-language model with a lightweight trainable projector bolted onto a large decoder, rather than a conventional small-parameter SLM.

7. Audio Model Comparison

Model	Approx. Size / Architecture	Output	Indian Languages
IndicConformer-600M	~600M	Text	22 Indian languages
Saaras	Not positioned as an SLM	Text / translated text	22 Indian + English (23 total)
Shuka v1	Llama 3 8B decoder + ~60M projector	Text / language response	Tested on Hindi, Kannada, Malayalam, Tamil, Telugu and others

8. Two Candidate System Architectures

Architecture 1 — Two-Stage Pipeline

Speech → IndicConformer / Saaras → Text → Sarvam 2B / Qwen / Gemma → MASSIVE intent

This is the conventional approach, and the one closest to the Whisper → MuRIL pipeline already in place. It keeps speech recognition and intent understanding as separate, independently optimizable components: a text SLM can be fine-tuned directly on MASSIVE, and different ASR models can be swapped in without touching the intent model.

Architecture 2 — Single Audio-Language Model

Speech → Audio Language Model → Language Understanding → Intent

This second direction asks whether a single model can go from audio straight to intent without a separate text-recognition stage. Shuka v1 is the example of this architecture in the current survey, though its 8B-parameter Llama decoder means it does not deliver the same efficiency benefits as the two-stage design — it is more useful here as a research direction than as a lightweight deployment candidate.

9. Model Groups Identified for Further Investigation

Group	Model(s)	Primary Purpose
A — Indian-language text SLM	Sarvam 2B	Indian-language text understanding and intent classification
B — General multilingual text SLMs	Qwen2.5, Gemma, SmoLM2, Llama 3.2	Baseline comparison for multilingual Indian-language intent classification
C — Small Indian-language speech model	IndicConformer-600M	Speech-to-text for the five target Indian languages
D — Indian-language speech model	Saaras	High-coverage Indian-language speech recognition and transcription
E — Audio-language model	Shuka v1	Investigating direct audio-to-language understanding

10. Key Observations

- An SLM does not have to process audio itself — a two-stage design (small ASR model feeding a small text SLM) is a practical baseline that mirrors the Whisper → MuRIL setup already in use.
- Sarvam 2B is the only candidate that is India-focused rather than generally multilingual; Qwen2.5, Gemma, SmoLM2 and Llama 3.2 are useful baselines, but their Indian-language performance is unverified until tested on the pooled MASSIVE subset.
- IndicConformer-600M is the strongest small-ASR candidate given its explicit support for all five target languages at roughly 600M parameters.
- Shuka v1 sits outside the “small model” framing despite its name recognition in this space — its Llama 3 8B decoder means any efficiency claims about it need that caveat attached.
- Grouping the ten candidates into Groups A–E (Section 9) gives a structure for deciding which models to actually fine-tune or evaluate against the Week 5 MASSIVE Indian-language subset.

11. Notes and Constraints

- None of the models above have been benchmarked yet on the Indian-language MASSIVE subset from Week 5 — the comparisons in this report are at the specification and architecture level only.
- Parameter counts are approximate, as published by each model's developer; actual memory and compute footprints will depend on quantization and serving setup, which this report does not account for.
- “Multilingual” is not the same claim as “Indian-language-optimized.” Qwen2.5, Gemma, SmoLM2 and Llama 3.2 are included as general baselines precisely because their Indian-language support has not been confirmed, not because it has.
- Shuka v1's zero-shot performance on languages beyond English and Hindi comes from its multilingual audio encoder rather than targeted fine-tuning, which is a meaningfully different guarantee than the other models' stated language support.